DEGRACE L1024 production outputs

Snapshot schedule and lens planes: what to implement in MG-GLAM

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16 October 2026 plan for review, nothing implemented yet

Agreed on Slack: lens planes instead of particles

Drop particle snapshots

2048^3 particles with positions and velocities: 200 GB per snapshot, 9 TB per box. Not feasible for 100 LCDM boxes.

Store projected density: lens planes

At every output, project the density along each of the 3 axes in 4 slabs of depth $256\,h^{-1}\mathrm{Mpc}$: **12 planes of** 4096^2 **float32**.

- `iSave=0`; checkpoints still written for restart, deleted at the end.
- Velocities survive only in the BDM halo catalogues (`CatshortV`).
- Scope: **L1024, Np2048, Ng4096**, fiducial Planck 2015, later the design cosmologies. Not the L1800 suite (256 does not divide 1800).

0.8 GB per snapshot (12 planes)

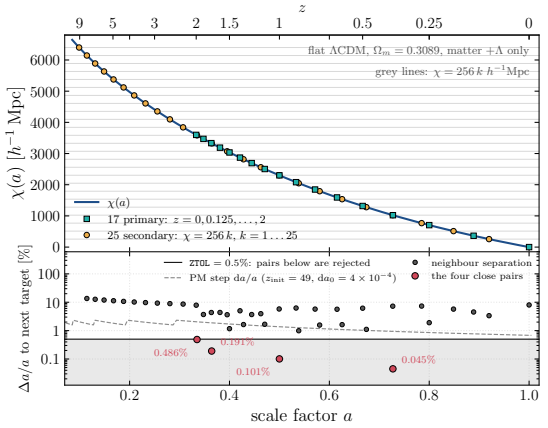
34 GB per box (42 outputs)

3.4 TB for 100 LCDM boxes

42 outputs per box $=$ 17 *primary* z $(0, 0.125, \ldots, 2$; common to all cosmologies$)$ $+$ 25 *secondary* at $\chi = 256\,k\,h^{-1}\mathrm{Mpc}$, $k = 1 \ldots 25$ (FORGE / DEGRACE-ECOSMOG convention).

Source: `slack-chat-degrace-prep.txt`; plan in `docs/degrace_companion_implementation_plan.md`

The 42-output schedule



Primary (17)

$z = 0$ to 2 in steps of 0.125. Same for every cosmology; the natural epochs for halo catalogues and emulator training.

Secondary (25)

$\chi(a) = 256\,k\,h^{-1}\mathrm{Mpc}$, $k = 1 \ldots 25$, out to $6400\,h^{-1}\mathrm{Mpc}$ ($z \simeq 9.1$). Cosmology dependent: recompute per design node.

Four close pairs

Separations 0.49%, 0.19%, 0.10%, 0.045% in a. The exact-z code rejects anything below 0.5% (slide I1).

Source: `figs/make_output_schedule_fig.py`; PM ladder for $z_{\mathrm{init}} = 49$, $\mathrm{d}a_0 = 4 \times 10^{-4}$

Cross-check: Baoju's scale factors are correct

Baoju asked for an independent check of his 42 values against the Planck 2015 parameters. Each secondary target was recomputed as the root of $\chi(a) = 256\,k\,h^{-1}\mathrm{Mpc}$ with $\chi(a) = \dfrac{c}{H_0}\displaystyle\int_a^1 \dfrac{\mathrm{d}a'}{a'^2 E(a')}$, $c/H_0 = 2997.92458\,h^{-1}\mathrm{Mpc}$.

background assumed	$E^2(a)$	$\max\lvert\Delta a/a\rvert$	$\Delta\chi$ at $6400\,h^{-1}\mathrm{Mpc}$
Planck 2015, matter $+\Lambda$ only	$0.3089\,a^{-3} + 0.6911$	7.7×10^{-8}	$< 0.001\,h^{-1}\mathrm{Mpc}$
plus radiation ($h = 0.6774$, $N_{\mathrm{eff}} = 3.046$)	$+\,\Omega_r a^{-4}$	1.7×10^{-3}	$2.8\,h^{-1}\mathrm{Mpc}$
$\Omega_m = 0.3156$ instead		2.9×10^{-2}	$49\,h^{-1}\mathrm{Mpc}$

Conclusions

- The list reproduces flat ΛCDM with $\Omega_m = 0.3089$ to his integrator's noise.
- Distance convention is matter $+\Lambda$ only; the lightcone code must use the same $\chi(a)$.

Do not reuse the Node_082 list

The 43-entry list read from bl267's ECOSMOG run is for the Node_082 cosmology (stations differ by up to 3% in a). `ic2pm_val_L1024` used it in legacy nearest-step mode, including $z_{\mathrm{init}} = 49$.

Source: scipy quad/brentq, cosemu env, 2026-10-16; snippet in the markdown plan

I1 (blocker): four pairs below the 0.5% tolerance

pair (z)	a	sep. in a	Δa
2.0000 / 1.98541	0.33333 / 0.33496	0.486%	1.6×10^{-3}
1.7500 / 1.74475	0.36364 / 0.36433	0.191%	7.0×10^{-4}
1.00202 / 1.0000	0.49950 / 0.50000	0.101%	5.1×10^{-4}
0.37562 / 0.3750	0.72695 / 0.72727	0.045%	3.3×10^{-4}

`SetExactSteps`: a target within `ZTOL`$= 0.5\%$ of a scheduled step *moves* that step onto the target; two targets closer than `ZTOL` are rejected, because the second would re-move the step pinned to the first. Other neighbours are $\geq 0.99\%$ apart; the late-time PM step is 6.8×10^{-3}, so the closest pair is $\sim 1/20$ of a step.

Tests (`test/precise_z/`, Nrow 128)

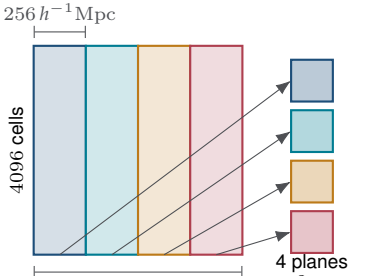
The four pairs $\rightarrow$ 42 marked steps, headers bit-exact at $1/(1+z)$, no duplicate step, one restart across a pinned pair; existing cases unchanged.

Proposed fix

1. Never move a step that is already pinned (`MarkX=1`); insert a new step instead. Keep 0.5% for moving *unpinned* steps, so 35 of the 42 targets still merge and only $\sim$7 PM steps are added.
2. Relax the pair rejection to "distinct in Real*4" (relative separation $> 10^{-5}$); keep the error for true duplicates.
3. Post-conditions: all $\mathrm{d}a > 0$, `Alist` strictly increasing, exactly `Nexact` pinned steps, each at the Real*4 of $1/(1+z)$.
4. Restart check unchanged (checkpoint a must equal `Alist(ISTEP)`).
5. The variable-step leapfrog already handles the short step ($F_V \approx 0.09$ then ≈ 1.9 around $\Delta a = 3.3 \times 10^{-4}$).

Source: `PMP2main.f90` `SetExactSteps`; 244-step ladder for $z_{\rm init} = 49$, $\mathrm{d}a_0 = 4 \times 10^{-4}$

I2: lens-plane output



Algorithm (generalises `ProjectDENSIT`)

- After `DENSIT` and **before** `GetPower`, which overwrites the mesh `FI`.
- Per axis: slab $s = \lfloor (i-1)/(N_{\rm grid}/N_{\rm slab}) \rfloor + 1$; sum $1+\delta$ over the slab's cells of each column, divide by their number $\Rightarrow$ mean $1+\delta$ of the slab column ($\Sigma/\bar{\Sigma}$).
- OpenMP over the outer loop; extra memory $12 \times 4096^2 \times 4\,\mathrm{B} = 0.8\,\mathrm{GB}$.
- Logged checks: each plane averages to 1; the 4 slabs sum to the full-box projection.

Parameters

Two lines *appended* to `Init.dat/Setup.dat`, read with `iostat`, default off: slabs per axis (4) and plane grid (4096, must divide $N_{\rm grid}$). Every existing run directory stays valid.

File `LensPlanes.<step>.<box>.dat`, one per output

Stream access, explicit little-endian (the build is globally big-endian). Header: a, step, $N_{\rm grid}$, $N_{\rm lens}$, $N_{\rm slab}$, Box, depth, Ω_m, Ω_Λ, h. Then the 12 planes in fixed order $\mathrm{X}_{1..4}$, $\mathrm{Y}_{1..4}$, $\mathrm{Z}_{1..4}$, each $N_{\rm lens}^2$ real*4, column-major. A numpy reader goes into halocat.

Source: `PMP2mod_density.f90` `ProjectDENSIT`; `PMP2mod_analyze.f90` `Analysis`; `PMP2init.f90`; `PMP2mod_tools.f90`

I3–I6: schedule, index file, content control, config

I3 Output schedule

- `Init.dat`: `#outputs = -42` + the 42 redshifts (exact mode).
- **Do not list** $z_{\rm init} = 49$: exact mode enforces $0 \le z < z_{\rm init}$; the IC file is the $z = 49$ snapshot.
- Generator script: stations from $\chi(a)$ of the run's Ω_m, merged with the 17 primary z; must reproduce Baojiu's list to $< 10^{-7}$.

I4 Step $\rightarrow$ redshift index

Snapshot files are named by `ISTEP`, irregular once targets are inserted. Write `OutputSchedule.dat` at run start (`ISTEP a z chi exact`); halocat loaders map snapshot numbers from it.

I5 Per-output content control (only if needed)

`iBDM`, `iPower` and the lens-plane flag are global: they fire at every marked step. If halo catalogues are wanted only at the 17 primary redshifts, add a per-redshift class consulted by the output block in `PMP2main.f90`. Deferred until Baojiu answers question 1.

I6 Run configuration and provenance

- `iSave=0`, `iPower=1`, `iBDM=1`, lens planes on.
- `Ncheckpoint` small enough that losing a 72 h job does not restart from $z_{\rm init}$ ($\sim$250 PM steps per box).
- Per box: `Init.dat`, `OutputSchedule.dat`, the 2LPTIC/`ic2pm` velocity-epoch line (`AGENTS.md`).

Source: `PMP2init.f90` parser; limits `zout(1000)`, `NstepM=2000`

Disk budget per box and open questions for Baojiu

product	per snapshot	per box	100 boxes
lens planes (12×4096^2 float32)	0.8 GB	34 GB	3.4 TB
`CatshortV` catalogues†	0.3–0.4 GB	∼15 GB	∼1.5 TB
`PowerDM`	kB	–	–
particle snapshots (`iSave=1`)	200 GB	8.4 TB	no
`PMcrs` checkpoints (transient)	200 GB	200 GB	deleted

† pilot model65 box 1, `MassMin`$= 5 \times 10^{11}$, if BDM runs at all 42 outputs.

Open questions

1. BDM catalogues at all 42 outputs, or only at the 17 primary redshifts?
2. Lens planes at the primary redshifts too, or only at the 26 stations ($z = 0$ and $k = 1 \ldots 25$)? His 34 GB assumes all 42.
3. Plane normalisation: mean $1 + \delta$ of the slab column (proposed) or surface density? Must match his ECOSMOG planes.
4. Keep the four close pairs (fix I1) or drop one of each? They differ by $\leq 1.5\,h^{-1}\mathrm{Mpc}$ in χ.

Source: pilot model65 box 1, `Run1/CATALOGS`

Summary and next steps

Done

- Baojiu's 42 scale factors verified against Planck 2015 (matter $+\Lambda$).
- Existing code located: exact-z scheduler, `ProjectDENSIT`, output block.
- Plan written: `docs/degrace_lensplanes_implementation_plan.md`.

Blocker

I1: the 0.5% pair rejection in `SetExactSteps` refuses Baojiu's list as is.

Order of work (branch off `cz`)

1. Reply to Baojiu: cross-check result, questions 1–4.
2. I1 pinned-step fix + `test/precise_z` cases.
3. I2 lens planes: routine, parameters, file format, numpy reader.
4. I3 schedule generator; I4 `OutputSchedule.dat`.
5. I5 only if question 1 says "primary only".
6. Pilot one L1024 box with the full 42-output schedule; check efficiency and sizes before the 100-box campaign.

Everything above is a proposal for review; no source has been changed.